certificateValidation

Service Description

Service ID: *"certificateValidation"*

Abstract
This document describes the `certificateValidation` service, version 1.0.0, produced by the CertificateAuthority support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `certificateValidation` service, which provides checking certificates and trusted keys against the CA's register, and publishing the certificate revocation list. It is produced by the CertificateAuthority system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 Certificate Authority validation endpoints; the CRL is new.

2.2 How This Service Is Meant to Be Used

Systems check a presented certificate with `check-certificate` (status `GOOD`, `REVOKED`, `EXPIRED` or `UNKNOWN`) and a public key with `check-trusted-key`. Brokers, the relay and the Java systems use the CRL, which is also written to `CRL_DIR/ca.crl` at start, after every revocation and periodically.

2.3 Important Delimitations and Dependencies

Only certificates of this cloud CA can be `GOOD`.

3 Service Interface

This section describes the interfaces to the `certificateValidation` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `certificateValidation` service and its operations.

The following interface operations are available.

3.1 operation check-certificate (CertificateCheckRequest) : CertificateCheckResult

Returns the status of a certificate.

3.2 operation check-trusted-key (TrustedKeyCheckRequest) : TrustedKeyCheckResult

Returns whether a public key is trusted now.

3.3 operation get-crl ()

Returns the current CRL (every revoked, unexpired certificate; next update after `CRL_VALIDITY_HOURS`, 168).

4 Information Model

Here, all data objects that can be part of the `certificateValidation` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.6, which are used to represent things like hashes and identifiers.

4.1 struct CertificateCheckRequest

Field	Type	Description
certificate	String	Certificate to check, PEM or Base64 DER (alias `encodedCertificate`).

4.2 struct CertificateCheckResult

Field	Type	Description
status	String	`GOOD`, `REVOKED`, `EXPIRED` or `UNKNOWN` (not issued by this CA).
valid	Boolean	True only for `GOOD`.
serialNumber	String	Serial number, decimal.
known	Boolean	Whether the serial is in the register (only for certificates of this CA).

4.3 struct TrustedKeyCheckRequest

Field	Type	Description
publicKey	String	PEM SubjectPublicKeyInfo (alias `publicKeyPem`).

4.4 struct TrustedKeyCheckResult

Field	Type	Description
trusted	Boolean	Whether the key is a trusted key within its validity.
key	TrustedKeyRecord	The matching record, or null.

4.5 struct TrustedKeyRecord

Field	Type	Description
id	Number	Record identifier.
public_key_pem	String	The key, PEM.
description	String	Description.
valid_after	String	Start of validity.
valid_before	String	End of validity.
created_at	String	Creation time.

4.6 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “CertificateAuthority – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-CertificateAuthority-java-spring/documentation/v5.3.0/CertificateAuthority-v5.3.0-SysD.pdf.

[2] ——, “certificateValidation HTTP/TLS/JSON – Interface Design Description (CertificateAuthority),” Eclipse Arrowhead, IDD 5.3.0, 2026, certificateAuthority-v5.3.0-certificateValidation-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	